

Math for America minicourse

Introduction to Category Theory

Session 2: Sameness

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Wednesday May 10, 2023

Plan

1. Recap
2. More examples of categories: sets, ordered sets, groups
3. Invertibility
4. Isomorphisms of sets
5. Isomorphisms of groups
6. Isomorphisms of ordered sets

1. Recap

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- We thought about how pure math comes from abstractions and analogies.
- We thought about properties of equivalence relations.
- We saw the definition of category, generalizing equivalence relations.
- We looked at some examples including numbers, factors, and privilege.

This week:

Today we'll look at some more examples,
and see how the framework of categories gives us a more nuanced way to talk about sameness.

1. Recap of what category theory is for

Category theory is the “mathematics of mathematics”

Why do we even do math at all?

- To solve problems.
- It's fun and interesting.
- We can apply it to other parts of life.
- It helps us understand the world better.
- It helps us make connections between different things in the world.

Category theory does all these things for us in math and therefore also life.

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 $a \longrightarrow b$

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an identity arrow $a \xrightarrow{1_a} a$
- **composition**: given $a \xrightarrow{f} b \xrightarrow{g} c$
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Properties (axioms)

- **unit**: given $a \xrightarrow{f} b$

$$\begin{aligned} a \xrightarrow{1_a} a \xrightarrow{f} b &= a \xrightarrow{f} b \\ a \xrightarrow{f} b \xrightarrow{1_b} b &= a \xrightarrow{f} b \end{aligned}$$

- **associativity**: given $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$

$$(h \circ g) \circ f = h \circ (g \circ f)$$

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$$(h \circ g) \circ f = h \circ (g \circ f)$$

What happened to symmetry?

We don't demand it but we look for it afterwards.
This is the notion of "sameness" in a category.

1. Recap: Examples we saw

1.
 - objects: natural numbers $0, 1, 2, \dots$
 - morphisms: $a \longrightarrow b$ whenever $a \leq b$
2.
 - objects: factors of 30
 - morphisms: $a \longrightarrow b$ whenever a is a multiple of b
3.
 - objects: factors of n
 - morphisms: $a \longrightarrow b$ whenever a is a multiple of b
4.
 - objects: a shape (just one object)
 - morphisms: symmetries of the shape
5.
 - objects: subsets of $\{2, 3, 5\}$ or $\{a, b, c\}$ or $\{\text{rich, white, male}\}$
 - arrows: subset relationships

2. More examples: Sets and functions

There is a large category with

- objects: all possible sets
- arrows: all possible functions

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Functions have a fixed domain and codomain (range).

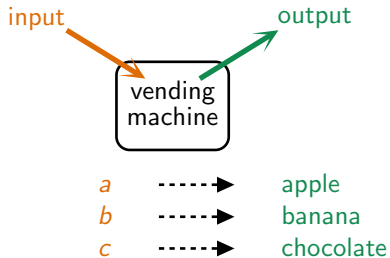
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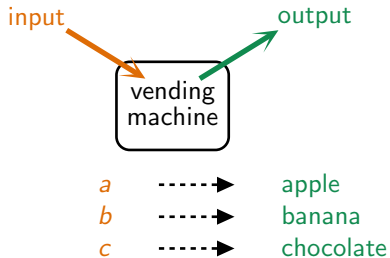
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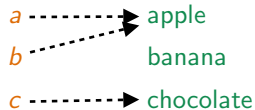
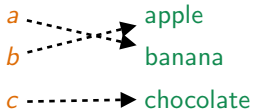
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But we could do less “sensible” things:



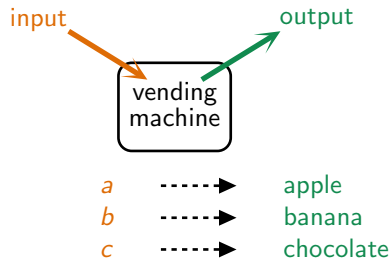
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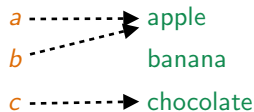
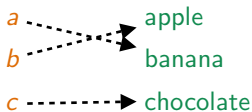
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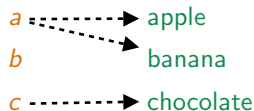
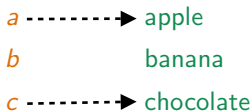
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Every input has to produce exactly one thing so these don't count.

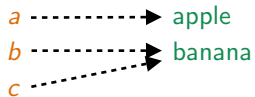


In this category:

a morphism $A \longrightarrow B$ is a function $A \longrightarrow B$.

2. More examples: order-preserving functions

In some cases the lines don't have to cross



What other possibilities are there?

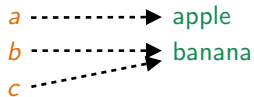
a apple
b banana
c

a apple
b banana
c

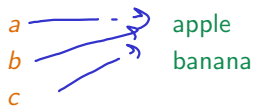
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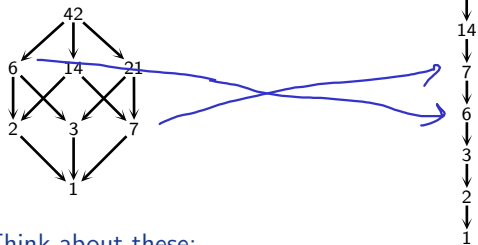
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When lines don't cross this is called order-preserving.



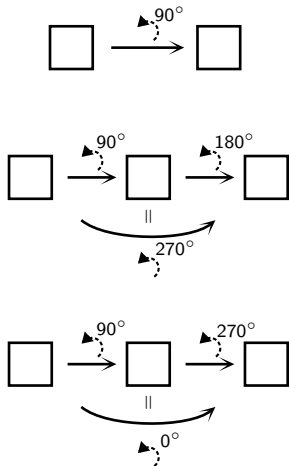
Think about these:

$\{\text{people, age}\} \rightarrow \{\text{people, wealth}\}$
 $\{\text{people, privilege}\} \rightarrow \{\text{people, wealth}\}$

There is a category of ordered sets and order-preserving functions

2. More examples: Sets with structure

We regarded symmetry as a relation and produced the table on the right



	0	90	180	270
0	0	90	180	270
90	90	180	270	0
180	180	270	0	90
270	270	0	90	180

The table gives a **binary operation** on the elements.

This is called a group. It has to satisfy some axioms:

- associativity
- there is an identity element which “does nothing”
- every element has an inverse which “undoes” it

This is a category with a single object.

2. More examples: Zooming in and out

Categories work at different scales.

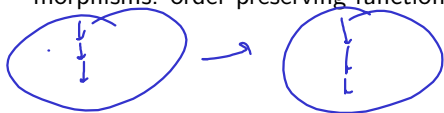
Zoom in: an ordered set is a category.

- objects: elements of the set
- morphisms: $\leq$



Zoom out: there is a large category of ordered sets.

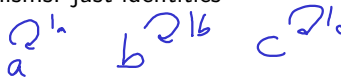
- objects: ordered sets
- morphisms: order-preserving functions



We can do this on sets themselves.

Zoom in: a set is a category.

- objects: elements of the set
- morphisms: just identities



Zoom out: there is a large category ^{of} sets

- objects: sets
- morphisms: functions



Categories are a generalization of sets.

2. More examples: Zooming in and out

Zoom in: the group of symmetries of a square is a category

- objects: one single object, the square
- morphism: the symmetries

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This generalizes to other shapes.

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Zoom out: there is a large category of groups

- objects: groups
- morphisms: structure-preserving functions

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Zoom in: the group of symmetries of a square is a category

- objects: one single object, the square
- morphism: the symmetries

This generalizes to other shapes.

Zoom out: there is a large category of groups

- objects: groups
- morphisms: structure-preserving functions

Technicalities:

- When we did ordered sets we also had structure-preserving morphisms.
- There the structure was the ordering, so the morphisms had to preserve the order.
- Here the structure is the binary operation, so the morphisms have to preserve that.
- This means

$$f(a \circ b) = f(a) \circ f(b)$$

We'll come back to this later.

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3. Invertibility

In category theory we try and express things just using relationships.

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For example, can we say these sets are “the same” without mentioning the elements?

$$\{1, 2, 3\} \equiv \{a, b, c\}$$

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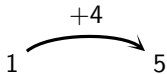
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We use **invertibility** of morphisms.

This is about going back to where you started or undoing a process.



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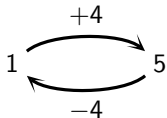
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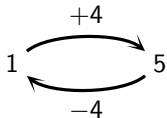
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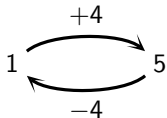
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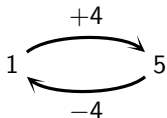
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Are these processes undoable?

i. $\xrightarrow{+4}$

ii. $\xrightarrow{\times 4}$

iii. $\xrightarrow{+0}$

iv. $\xrightarrow{\times 0}$

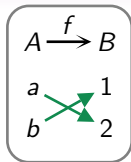
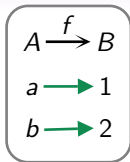
v. $\xrightarrow{\text{squaring}}$

vi. All the functions $\{a, b\} \longrightarrow \{1, 2\}$. Which do you think should count as invertible?

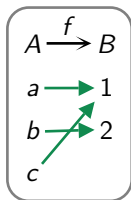
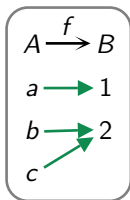
vii. What about functions $\{a, b, c\} \longrightarrow \{1, 2\}$?

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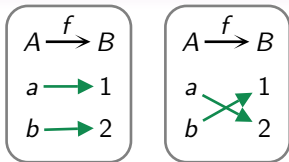


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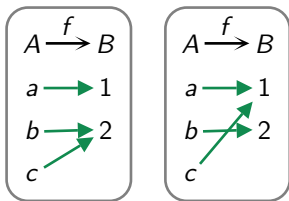


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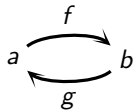
- In what sense is/isn't divorce the inverse of marriage?
- In what sense is/isn't a pardon the inverse of a criminal conviction?

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Definition of inverses in category theory

15.

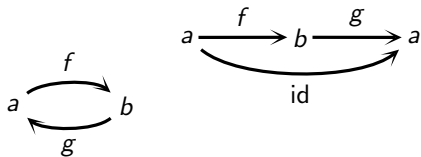
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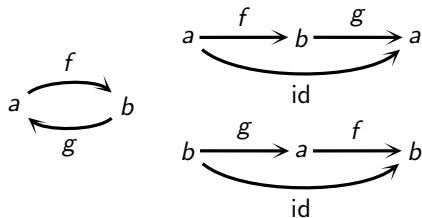
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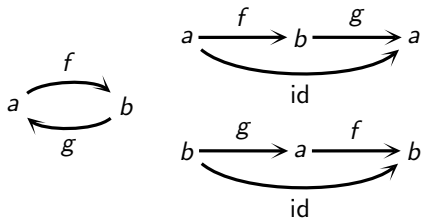


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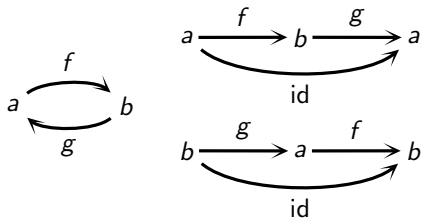
In that case

- f and g are inverses of each other
- they are called invertible, also isomorphisms
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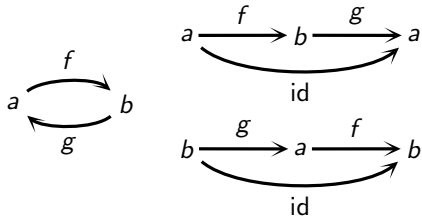
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Isomorphic objects are treated as the same by the rest of the category.

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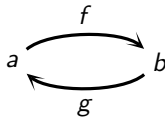
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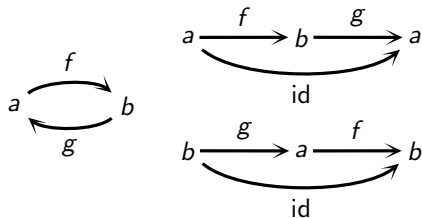
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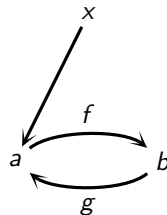
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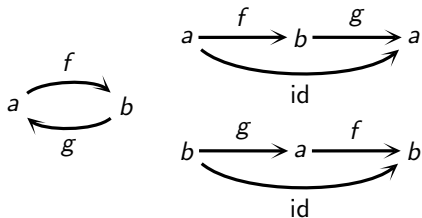
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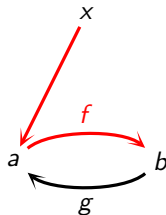
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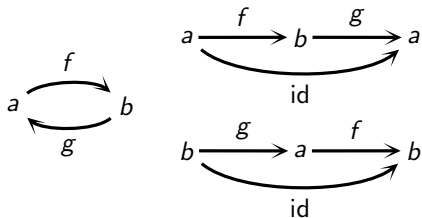
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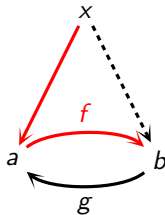
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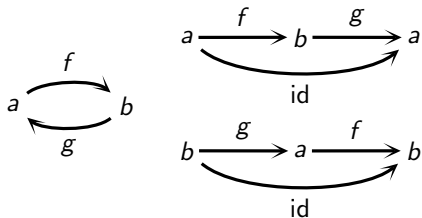
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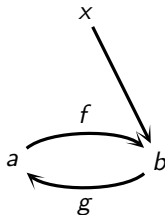
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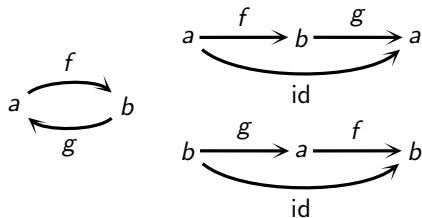
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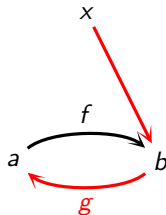
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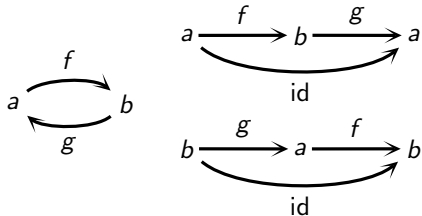
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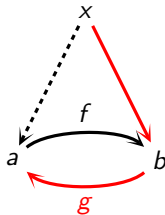
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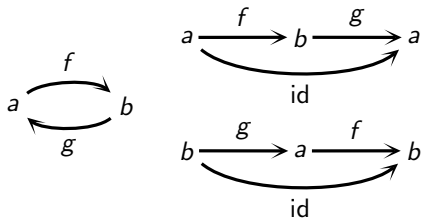
Isomorphic objects are treated as the same by the rest of the category.



3. Invertibility

Definition of inverses in category theory

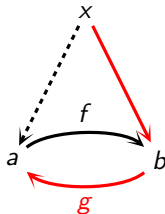
An inverse for f is g such that:



In that case

- f and g are inverses of each other
- they are called invertible, also isomorphisms
- a and b are called isomorphic

Isomorphic objects are treated as the same by the rest of the category.



Things to try proving:

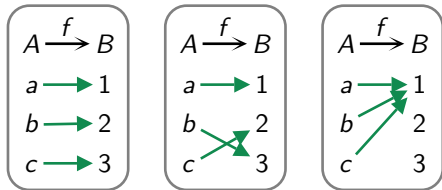
- Inverses are unique.
- Isomorphism is an equivalence relation.

Break to 6:35

4. Isomorphisms of sets

When is a function an isomorphism?

16.

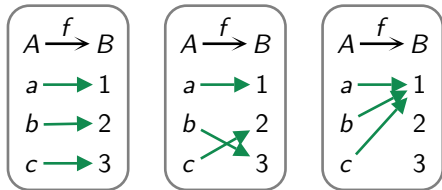


Number of isomorphisms is:

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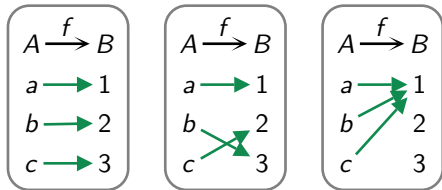


Number of isomorphisms is: $3 \times 2 \times 1 = 6$

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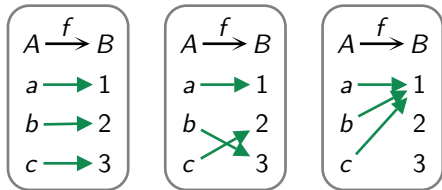
Similar to permutations!

Isomorphisms $A \longrightarrow B$ are bijections.

- For finite sets: same number of elements.
- For infinite sets: same infinity. . .

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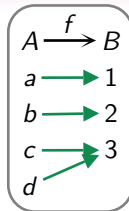


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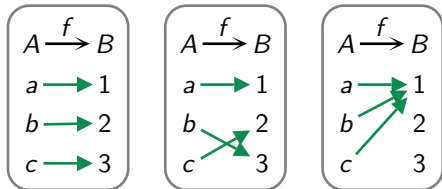
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Why, technically, is f not an isomorphism?

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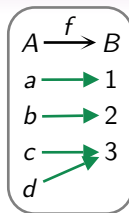


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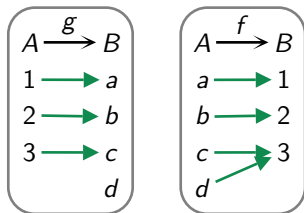
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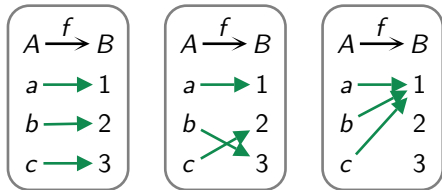
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We can try to define an inverse g .



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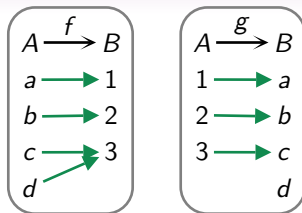


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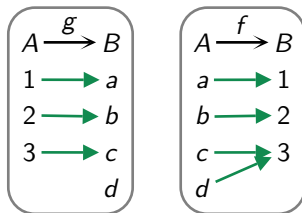
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We can try to define an inverse g .



5. Isomorphisms of groups

A group is a set with a binary operation satisfying associativity, identities, and inverses.

- It is a category with one object in which every morphism is an isomorphism.
- But we can zoom out: what is an isomorphism of groups?

Key: patterns in tables

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Key: patterns in tables

Rotations of square $\equiv$ addition on 4-hour clock

	0	90	180	270
0	0	90	180	270
90	90	180	270	0
180	180	270	0	90
270	270	0	90	180

	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

These patterns are “the same”.

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0	0	90	180	270
90	90	180	270	0
180	180	270	0	90
270	270	0	90	180

	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

These patterns are “the same”.

Try $\mathbb{Z}_{10}$ (“10-hour clock”) $\times$

	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

	1	3	9	7
1	1	3	9	7
3	3	9	7	1
9	9	7	1	3
7	7	1	3	9

We have to re-order this to see the pattern.

Try $\mathbb{Z}_8$ (8-hour clock) $\times$.

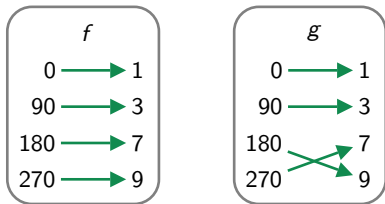
What pattern do you see?

	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

5. Isomorphisms of groups

Here are two functions from the rotations of a square to $\{1, 3, 9, 7\}$ in $\mathbb{Z}_{10}$

18.

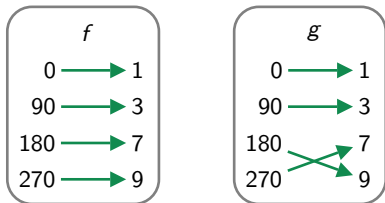


Which do you think should count as “pattern preserving” and which not?

	0	90	180	270		1	3	7	9		1	3	9	7
0	0	90	180	270	1	1	3	7	9	1	1	3	9	7
90	90	180	270	0	3	3	9	1	7	3	3	9	7	1
180	180	270	0	90	7	7	1	9	3	9	9	7	1	3
270	270	0	90	180	9	9	7	3	1	7	7	9	3	9

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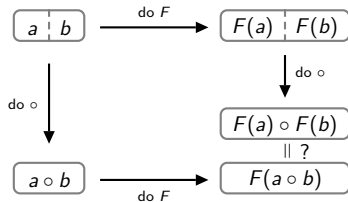


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	0	90	180	270		1	3	7	9		1	3	9	7
0	0	90	180	270	1	1	3	7	9	1	1	3	9	7
90	90	180	270	0	3	3	9	1	7	3	3	9	7	1
180	180	270	0	90	7	7	1	9	3	9	9	7	1	3
270	270	0	90	180	9	9	7	3	1	7	7	3	9	9

Preserving pattern is about respecting $\circ$

A **group homomorphism** $G \xrightarrow{f} H$ is a function such that for all $a, b \in G$, $f(a \circ b) = f(a) \circ f(b)$



Groups and homomorphisms form a category.

5. Isomorphisms of groups

- Isomorphisms in the category of groups and group homomorphisms are group homomorphisms with an inverse.
- This turns out to mean they have the same pattern.
- There is only one possible pattern for a group of 2 elements. We say there is only one group with 2 elements “up to isomorphism”.
- There is also only one group with 3 elements “up to isomorphism”.



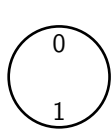
For example addition on a “3-hour clock” (integers modulo 3).

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

This would also be the same pattern as rotations of an equilateral triangle.

5. Isomorphisms of groups

Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

20.

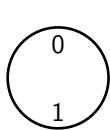
Battenberg Cake



×	0	1
0		
1		

5. Isomorphisms of groups

Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

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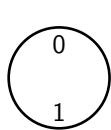
Battenberg Cake



×	0	1
0	0	0
1	0	1

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Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

×	even	odd
even	even	even
odd	even	odd

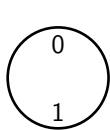
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×	0	1
0	0	0
1	0	1

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Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

×	even	odd
even	even	even
odd	even	odd

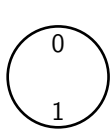
Battenberg Cake



×	0	1
0	0	0
1	0	1

5. Isomorphisms of groups

Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

Battenberg Cake



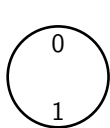
×	0	1
0	0	0
1	0	1

×	even	odd
even	even	even
odd	even	odd

×	real	imaginary
real	real	imaginary
imaginary	imaginary	real

5. Isomorphisms of groups

Here are some examples of the 2-element group.



+	0	1
0	0	1
1	1	0

+	even	odd
even	even	odd
odd	odd	even

Battenberg Cake



×	0	1
0	0	0
1	0	1

×	even	odd
even	even	even
odd	even	odd

×	real	imaginary
real	real	imaginary
imaginary	imaginary	real

×	tolerant	intolerant
tolerant	tolerant	intolerant
intolerant	intolerant	tolerant

5. Isomorphisms of groups

- There are only two possible patterns for a group of 4 elements.
- We say there are only two groups with 4 elements, “up to isomorphism”.

We have seen the two possible patterns:

	0	90	180	270
0	0	90	180	270
90	90	180	270	0
180	180	270	0	90
270	270	0	90	180

$\mathbb{Z}_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

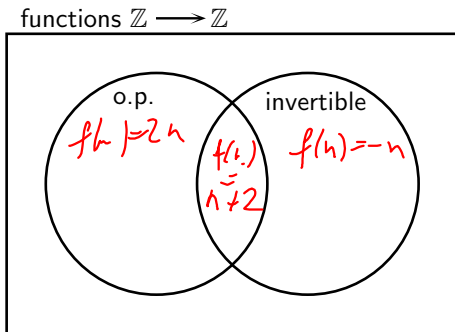
6. Isomorphisms of ordered sets

Isomorphisms of ordered sets are functions that are both order-preserving and invertible.

Consider these functions $\mathbb{Z} \longrightarrow \mathbb{Z}$.

Are they order-preserving? Invertible?

- i. $f(n) = 2n$
- ii. $f(n) = -n$
- iii. $f(n) = n + 2$
- iv. Can you figure out what *all* the (order-preserving) isomorphisms are? See if you can put things in areas of the Venn diagram.

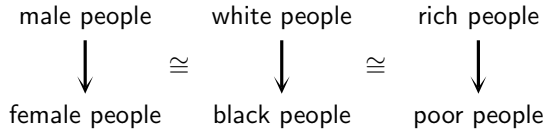


6. Isomorphisms of ordered sets

When we draw an ordered set like a category
an isomorphism shows “the same pattern” of arrows.

23.

These whole categories are isomorphic.

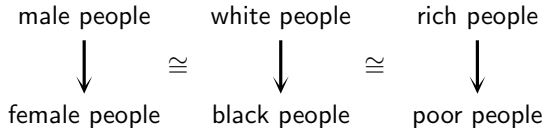


6. Isomorphisms of ordered sets

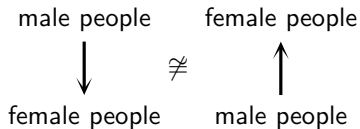
When we draw an ordered set like a category
an isomorphism shows “the same pattern” of arrows.

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These whole categories are isomorphic.



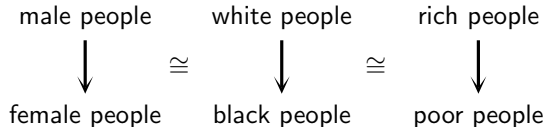
However:



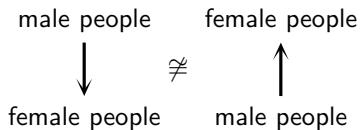
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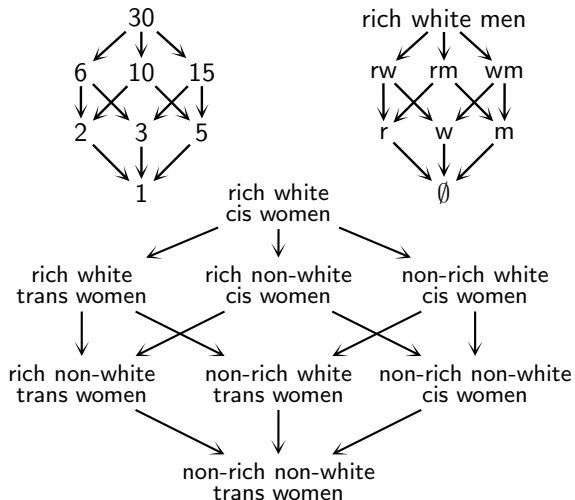
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However:



These are also isomorphic.



What equality really means

Equality in category theory is not about when things **are** the same,
but when the category **treats them** as the same.

What equality really means

Equality in category theory is not about when things **are** the same,
but when the category **treats them** as the same.

Equality in society should be about
society treating people as the same.